

Roll No.

--	--	--	--	--	--	--

ODD END SEMESTER EXAMINATION DECEMBER- 2025

Name of the Course: B.Tech (ME)

Semester: III

Name of the paper: Thermal Engineering

Paper Code: TME 308

Time: 3 hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1.	20 marks				
(a)	Define internal energy and prove that it is a property of a system.				CO1
(b)	A closed system of constant volume experiences a temperature rise of 25°C when a certain process occurs. The heat transferred in the process is 30 kJ. The specific heat at constant volume for the pure substance comprising the system is 1.2 kJ/kg°C, and the system contains 2.5 kg of this substance. Determine (i) The change in internal energy and (ii) The work done.				
(c)	Determine the dryness fraction and specific internal energy of steam at 7 bar and enthalpy 2550 kJ/kg. Relevant steam table extract is given below.				
	P (bar)	Specific enthalpy (kJ/kg)		Specific internal energy (kJ/kg)	
		h_f	h_{fg}	u_f	u_g
	7	697.1	2064.9	696	2573
Q2.	20 marks				
(a)	Explain turbine and compressor with suitable diagram. Derive the steady flow energy equation for both the devices.				CO2
(b)	In a gas turbine unit, the gases flow through the turbine is 15 kg/s and the power developed by the turbine is 12000 kW. The enthalpies of gases at the inlet and outlet are 1260 kJ/kg and 400 kJ/kg respectively, and the velocity of gases at the inlet and outlet are 50 m/s and 110 m/s respectively. Determine (i) The rate at which heat is rejected to the turbine, and (ii) The area of the inlet pipe given that the specific volume of the gases at the inlet is 0.45 m ³ /kg.				
(c)	Two Carnot engines work in series between the source and sink temperatures of 550 K and 350 K. If both engines develop equal power, determine the intermediate temperature.				
Q3.	20 marks				
(a)	Show that for an ideal gas, the slope of the constant volume line on the T-s diagram is more than that of the constant pressure line.				CO3
(b)	A mixture of ideal gases consists of 4 kg of nitrogen and 6 kg of carbon dioxide at a pressure of 4 bar and a temperature of 20°C. Determine (i) The mole fraction of each constituent, (ii) The equivalent molecular weight of the mixture, and (iii) The equivalent gas constant of the mixture.				

(c)	Derive the following expression: $U = F - T \left(\frac{\partial F}{\partial T} \right)_V$ Where U = Internal energy, F = Helmholtz function, T = Temperature, V = Volume	
Q4.	20 marks	
(a)	Describe Lenoir and Atkinson cycle with the help of p-v & T-s diagrams.	CO4,
(b)	In an air standard Diesel cycle, the compression ratio is 16, and at the beginning of isentropic compression, the temperature is 15°C and the pressure is 0.1 MPa. Heat is added until the temperature at the end of the constant pressure process is 1480°C. Determine (i) The cut-off ratio, (ii) The heat supplied per kg of air, and (iii) The cycle efficiency	CO5
(c)	In an engine working on Dual cycle, the temperature and pressure at the beginning of the cycle are 90°C and 1 bar respectively. The compression ratio is 9. The maximum pressure is limited to 68 bar and total heat supplied per kg of air is 1750 kJ. Determine (i) Pressure and temperatures at all the points and (ii) Air standard efficiency.	
Q5.	20 marks	
(a)	Describe the working principle of a reverse Brayton cycle with the help of a p-v and T-s diagram.	CO6
(b)	Describe the mechanism of a simple vapour compression system with schematic diagram.	
(c)	In a refrigeration plant working on reversed Brayton Cycle, air is compressed to 5 bar from 1 bar. Its initial temperature is 10°C. After compression, the air is cooled up to 20°C in a cooler before expanding back to a pressure of 1 bar. Determine the COP of the plant and net refrigerating effect. Take $C_p = 1.005$ kJ/kg K and $C_v = 0.718$ kJ/kg K.	